

Thirayu Meererksom

Technology Leader · AI Researcher · Digital Transformation Consultant

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PROFESSIONAL SUMMARY

Ph.D. researcher and full-stack technology leader with 15+ years bridging advanced technology (AI & software architecture) and business strategy (MBA). Proven record delivering national-scale AI healthcare platforms for Thai government agencies and enterprise solutions for global clients in Switzerland and the USA. Peer-reviewed author published in Scientific Reports (Nature portfolio).

EDUCATION

Ph.D. Candidate, Translational Medicine (International Program) 2018–Present

Suranaree University of Technology

Dissertation: Detection and Characterization of Focal Liver Lesions on CT Images Using Deep Learning. GPAX 3.72

Master of Business Administration (MBA) 2008–2010

Khon Kaen University

IS: Strategies for Improving the Production Cost of a Software House.

B.Sc. Computer Science 2004–2008

Khon Kaen University

Project: A Patient Navigation System in Hospital using RFID.

PROFESSIONAL EXPERIENCE

Biz & Tech Consultant Aug 2019–Present

KASIRA HOUSE Co., Ltd.

- Directed national-scale AI healthcare platforms (AI-OVCCA, AI-Malaria, AI-Snaily) across provinces with NECTEC & the Ministry of Public Health.
- Designed scalable microservices (Docker, Kubernetes, Kong) and led KH-RMS hospital & KH-School delivery on .NET Core 8 + Angular.
- CyberSec Consultant at Big Boom Asia Co., Ltd.

Researcher Aug 2019–Present

Parasitic Disease Research Center, SUT

- Researcher at PDRC (SUT)

Senior Technical Partner (Remote) 2009–Present

PEZZ AG, Switzerland

- Sole technical partner for 15+ years; modernized legacy systems to cloud SaaS and iOS/Android with ~99.9% uptime across Europe.

Lecturer & Special Project Lead 2011–2019

Nakhon Ratchasima Rajabhat University

- Owned the B.B.A. Business Computer curriculum; led university digital projects (e-Learning, MIS) and mentored developers.

Academic Staff (Educational Technologist) 2010–2011

Rajamangala University of Technology Isan

Project Manager 2009

Cimatt Business Group Co., Ltd.

.NET Senior Developer 2008–2009

Miracle Inspire Co., Ltd.

.NET Developer 2008

Genius Megasofts Co., Ltd.

CONSULTING & ADVISORY

Registered Thai Consultant 2024–Present

Public Debt Management Office (PDMO), Ministry of Finance — Thai Consultant Database

Registered consultant in the national Thai Consultant Database. · 04/06/2567 ที่ปรึกษาอิสระ หมายเลข 6481 ระดับ 3

SME Coach (in-depth consultant) Thai SME-GP [ISMED] Marketing strategy, government procurement, ISO, finance/funding. · Clients: Bioway, Dairy Home, K.P. Generation, S-Camp, Coffee Roasting, etc.	2022–Present
Speaker & in-depth consultant Thai SME-GP [SSRN: Isan + Northern] Gov procurement, MarTech, SME ONE ID, Google Business, Canva, photography. · ภาคอีสานและภาคเหนือ	2023–Present
Speaker & in-depth consultant Thai SME-GP [SSRU Isan] Gov procurement, MarTech, ISO/GMP/MIT, Google Business, Canva + continued MSME consulting. · ที่ปรึกษาต่อเนื่องแก่ MSME 10+ ราย	2024–Present
Freelance Business & Digital Consultant Independent Online marketing, import–export, packaging design. · Clients: Coffee Roasting, Chamoke	2022–2024
Business Consultant KASIRA HOUSE Co., Ltd. Digital transformation & business strategy.	2019–Present
Cybersecurity Consultant Big Boom Asia Co., Ltd. Information security advisory.	2019–Present
Expert Lab Verification (Helminths) Department of Disease Control (กองโรคติดต่อทั่วไป), MoPH Stool helminth verification by Formalin-Ether Concentration Technique · via Kasira House	2020–Present

INVITED SPEAKING & TRAINING

Student Readiness Program, Faculty of Public Health Vongchavalitkul University — Smart Public Health & Occupational Safety: Using Technology & AI Wisely	10 Sep 2025
7th National Local Scout Jamboree — Life Skills: E-Sport Khonburi School, Nakhon Ratchasima PAO — E-Sport skills training	6–10 Aug 2025
SME Restart Suan Sunandha Rajabhat University — Green Tourism for community-based tourism	Oct 2023
BDS Market Suan Sunandha Rajabhat University — SME Business Development Services (BDS) Market	Nov 2022–Jan 2026
Thai SME-GP Suan Sunandha Rajabhat University	Jan 2022–Jan 2026
AI for MSME Office of SMEs Promotion (OSMEP)	Oct 2022-Sep 2025
AI in healthcare Ministry of Public Health	Oct 2018-Sep 2024
Artificial Intelligence for Automated Microscopic Helminth Screening Department of Disease Control	Oct 2020-Present
ETDA DIGITAL CITIZEN (EDC) ETDA — ETDA Digital Citizen (EDC)	Jan 2022-Present
True Community Builder 2024 - 2025 TRUE Digital — True Digital Community Builder 2024 - 2025	Jan 2024-Dec 2025
OTOP Nawatwithi Community-Based Tourism (Nakhon Ratchasima and Buriram) Ministry of Social Development & Human Security	Jan 2019-Present
Workshop — Long Term Care (3C) & Blockchain health records Department of Health (Bureau of Elderly Health), MoPH — Care Manager / Caregiver (70/420 hr) curriculum & local elderly-care volunteers	17 Jan 2020
Career-readiness workshop (Digital Media program) Nakhon Ratchasima Rajabhat University — Job-hunting techniques & essential skills in the digital era	1 Apr 2025

Government & partner organizations served

Office of SMEs Promotion (OSMEP), Ministry of Public Health, Ministry of Higher Education, Science, Research & Innovation (MHESI), Ministry of Social Development & Human Security, Department of Disease Control, Department of Health, Thai Media Fund, ETDA, TRUE Digital, NECTEC, Ministry of Commerce (OTOP projects), Ministry of Interior — CDD (OTOP Village)

Selected training topics

- AI & Computer Vision (helminths, malaria, leprosy, snail detection)
- Deep Learning & NLP (e.g., human-trafficking surveillance)
- Digital & social-media marketing technology
- Line MyShop, Shopee, Lazada, TikTok Shop
- Mobile & web application development
- CMS: Joomla, Mambo, Moodle, WordPress
- OTOP product development & online marketing
- E-commerce for all (E&M Commerce)
- Business Model Canvas & related tools
- Content creation for content marketing
- Metaverse for education technology
- Smart tourism & geopark information (Khorat Geopark)
- MVC concept; UTP & basic networking
- Fleet management system for truck companies
- Rice co-operative logistics management
- ITPE knowledge for advanced IT
- Thailand World Skill — Micro-Computer Level 1
- E-Sport skills & competition
- Thai SME-GP and BDS Market

Documented engagements

- 17 Jan 2020 — Department of Health (กรมอนามัย, สำนักอนามัยผู้สูงอายุ) — Speaker — Long Term Care (3C)/Care Manager curriculum, Chiang Mai
- 1 Apr 2025 — Nakhon Ratchasima Rajabhat University (Digital Media) — Speaker — job-hunting & digital-era skills
- 10 Sep 2025 — Vongchavalitkul University (Public Health) — Speaker — Smart Public Health & Occupational Safety with AI
- 6–10 Aug 2025 — Khonburi School / Korat PAO — Trainer — E-Sport (Korat Esports VP)
- 31 Jan 2020 — Department of Disease Control (กองโรคติดต่อทั่วไป) — Helminth stool verification (Formalin-Ether) — Kasira House
- - — School board committee — Invitation — school board (Children's Day)

SELECTED PROJECTS & SYSTEMS**AI & Healthcare**

AI-Snaily — Real-time YOLO detection of liver-fluke host snails (98.7% accuracy) — basis of the 2026 Scientific Reports paper.

AI-OVMIF / AI-Helminths — Parasite-egg detection AI (OV, MIF, Cs, Ec, AI, Tt, Tspp) for DDC and NECTEC.

AI Malaria & AI Leprosy — Computer-vision diagnosis tools for DDC and Rajpracha Samasai Institute.

Anamai Chatbot & COVID-19 System — Rapid-response healthcare chatbot and pandemic info platform for the Dept. of Health.

SUT OVCCA / Kunlóm — Liver-fluke/CCA and elderly-fall self-risk screening tools.

3C & Long-Term Care — Web portal and GIS reporting system for the Dept. of Health.

Enterprise & FinTech

KH-RMS & KH-School — Hospital and school management platforms.

Secure Financial CRM & Dashboard — Mortgage/auto/credit tracking with bank-grade CI/CD (NDA).

ERP & Logistics + POS — Laravel + Angular ERP with SARIMA demand forecasting.

Algorithmic Trading & Automation — MT5/crypto trading bots and n8n/Make automation.

Global / PEZZ

GAS2008 / GAS2010 — Growth Analysis System for PEZZ AG, Switzerland.

Child-Growth / Growth-Guide / Bedwetting / PEZZPat — iOS/Android clinical apps for PEZZ AG.

Mobile & Web

Clear Mechanic (USA) — Android repair-order app over RESTful web services.

Khorat Geopark / Go Khorat — Cross-platform smart-tourism information for the UNESCO Global Geopark.

MegaDict / Thai Talk 2.0 / Pro Balancer — Desktop applications (C#, Speech API).

Enterprise & FinTech

KH-MICE — Korat MICE city platform (web & mobile)

KH-Form — Form builder beyond Google Forms, with Passkey auth

Korat Esport Match — Esports tournament data & challenge platform

Game Dashboard — JavaScript game analytics dashboard

Odoo API Integration — Odoo–WinSpeed integration via FastAPI

AI & Healthcare

AI-Teania — Parasite-egg (Tspp) detection AI

Malaria Online — Malaria exam & e-certificate platform

MG-OVCCA — OV-CCA risk-screening questionnaire (LINE LIFF)

Research & Systems

SUT-RDMS — Research Project Management System

SUT-IMPM — Laboratory & research-unit management system

ISAN Goat — Goat-farm research questionnaire (LINE LIFF)

AI & Healthcare

License Plate Recognition — AI license-plate recognition on edge devices

Community

TPA Career Path — Thai Programmer Association career-path roadmap

Enterprise & FinTech

ERP (PPCeramic) — POS & inventory ERP

Storm Manager — Developed a cost management and human resource management system that occurred in the USA, or Storm Manager in the form of a Web Application and Web Service, which includes the following features: • Event Recording System • Workforce Planning and Cost System • Map System • Accommodation Management System • Material and Equipment Management System • Financial System • Summary Report

ERP (Connect Diagnostics) — ERPNext Implementation for Medical Devices Company

ERP (World Fert) — ERP for Fertilize Wholesaler

RESEARCH PUBLICATIONS

ORCID 0000-0001-6414-3607

International Peer-Reviewed Journals

- Jenwithee T, Meerersom T, Limpanont Y, Sripa B, Laha T, Suwannatrat AT. (2026). Automated real-time surveillance of Bithynia snails using a comparative YOLO based approach for liver fluke host detection. Scientific Reports, 16, 14886. doi.org/10.1038/s41598-026-43387-x
- Kaewpitoon N, et al. (incl. Meerersom T). (2018). Detection of the carcinogenic liver fluke risk groups by verbal screening questionnaires using a mobile application. Asian Pacific Journal of Cancer Prevention.
- Kaewpitoon SJ, et al. (incl. Meerersom T). (2018). A carcinogenic human liver fluke infection among migrant workers in NE Thailand. Tropical Biomedicine, 35(2), 453–463.
- Kaewpitoon N, Kaewpitoon SJ, Meerersom T, et al. (2018). Detection of Opisthorchis viverrini infection among the ASEAN population in Thailand. Iranian Journal of Parasitology, 13(2), 258–266.

Conference Proceedings

- Meerersom T, Chan-aran S, Kaewpitoon SJ, Rujirakul R, Kaewpitoon N. (2016). SUT OVCCA Application: a high-potential tool for screening liver fluke and cholangiocarcinoma. 3rd Asian Symposium on Healthcare Without Borders, Fukuoka, Japan, pp. 90–96.
- Meerersom T. (2017). Health Application Development in Real-Life. 4th Asian Symposium on Healthcare Without Borders, Hiroshima, Japan, pp. 23–32.
- Meerersom T, Siripornmongkon W. (2018). Kunlóm: A self-risk assessment for falling among the elderly. 4th Conference on Public Health in Asia, Hiroshima, Japan.
- Kaewpitoon N, Meerersom T, et al. (2018). Effectiveness of the SUT-OVCCA mobile application for risk-group screening among rural populations. 5th Int. Congress on Infectious Diseases, Berlin, Germany.
- Kaewpitoon SJ, ..., Meerersom T, et al. (2018). Prevention and control of carcinogenic liver fluke in rural communities through an inter-/trans-disciplinary university. 5th Int. Congress on Infectious Diseases, Berlin, Germany.
- Kaewpitoon SJ, ..., Meerersom T. (2017). LaHanNa Project. 4th Engagement Thailand Annual Conference, Chulalongkorn University, Bangkok.
- Kaewpitoon N, Meerersom T, et al. (2017). A simple screening for liver-fluke risk group through a smartphone application. Annual Conference, Institute of Medicine, SUT.
- Meerersom T, Kaewpitoon SJ, Kaewpitoon N. (2016). SUT OVCCA iOS Application. National Conference, Institute of Medicine, SUT.

National Publications (Thai)

- ธีรยุทธ มิถุณษ์สม. (2555). การปฏิรูปการเรียนการสอนของคณะวิทยาการจัดการ มรท. นครราชสีมา โดยใช้การจัดการความรู้: กรณีศึกษาสาขาวิชาโครงสร้างข้อมูล.
- ธีรยุทธ มิถุณษ์สม. (2556). การศึกษาผลการจัดการเรียนรู้ด้วยเทคนิคการสอนแนะ รายวิชาสารสนเทศด้านการเงินและการธนาคาร.
- ธีรยุทธ มิถุณษ์สม. (2557). การพัฒนาผลสัมฤทธิ์ทางการเรียน วิชาการระบบสารสนเทศเพื่องานธุรกิจ ด้วยทฤษฎีการสร้างองค์ความรู้.
- ธีรยุทธ มิถุณษ์สม. (2558). การจัดการเรียนรู้ด้วยปัญหาเป็นฐาน รายวิชาความมั่นคงของระบบสารสนเทศ.
- ธีรยุทธ มิถุณษ์สม. (2558). การประเมินความพึงพอใจของประชาชนต่อ อบต.หนองค่าย อ.ประทาย จ.นครราชสีมา.
- ธีรยุทธ มิถุณษ์สม. (2559). การประเมินความพึงพอใจของประชาชนต่อ อบต.วังไม้แดง อ.ประทาย จ.นครราชสีมา.

TECHNICAL SKILLS

Frontend: HTML, CSS, JavaScript; Angular, React/Next.js, Vue/Nuxt.js, Webpack; Bootstrap, Tailwind, Material UI

Backend: .NET / .NET Core, PHP (Laravel), Python, Node.js/Express, Java

AI / Data: Computer Vision (YOLO), Deep Learning, NLP, Data Science, SARIMA forecasting, BI

Databases: MS SQL, MySQL, PostgreSQL, Oracle; MongoDB, CouchDB, Elasticsearch; Neo4j; RabbitMQ

Cloud / DevOps: Docker, Kubernetes, Kong API Gateway, NGINX; Azure, AWS, GCP; CI/CD; VMware

Mobile: Android (Java), iOS (Objective-C, Swift); Flutter/Dart, React Native, Ionic, Unity

Business / Other: Agile & Waterfall, BA/SA, UI/UX, ERP (Odoo, ERPNext), IoT/RFID, MarTech, SME Coaching

CERTIFICATIONS & LICENSES

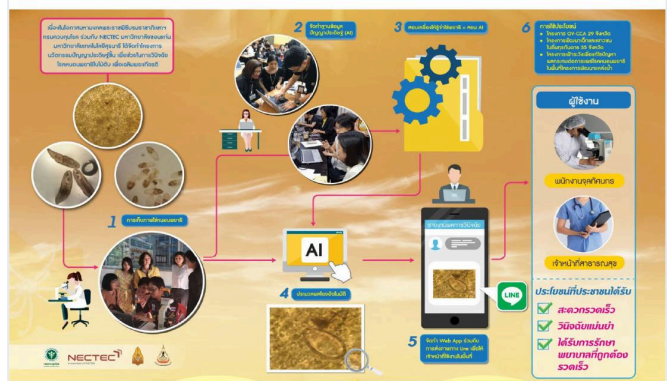
- Super AI Engineer, Level 1 (AIAT) [22p21i0089]
- Microsoft Certified Professional (MCP) & Application Developer (MCAD) [3622775]
- Sun Certified Java Programmer (SCJP), Java 2 Platform 1.4
- Information Technology Passport Examination (ITPE) [THIP15A0161]
- National Skill Standard — Micro-Computer Technician, Level 1 (Ministry of Labour) [บพฐ.1-4-21-001-0013-57]
- ThaiSP: Service Provider [SP57X95]
- Certified Thai SME Coach [8307]
- Public Debt Management Office (PDMO), Ministry of Finance — Thai Consultant Database [หมายเลข 6481 ระดับ 3]

PROFESSIONAL MEMBERSHIPS & LEADERSHIP

- **Vice President** — Korat Esports Sports Club, Nakhon Ratchasima
- **Committee Member** — Thai Programmer Association (TPA)
- **Member** — Artificial Intelligence Association of Thailand (AIAT)
- **Researcher** — Parasitic Disease Research Center, SUT



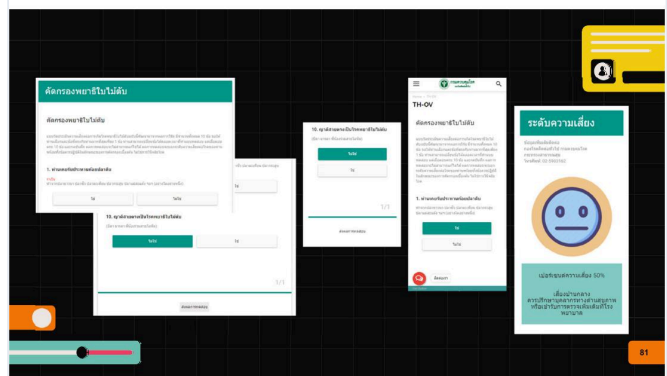
SUT-OVCCA: Liver-fluke/CCA self-risk screening tools and Kunlóm: elderly-fall self-risk screening tools.



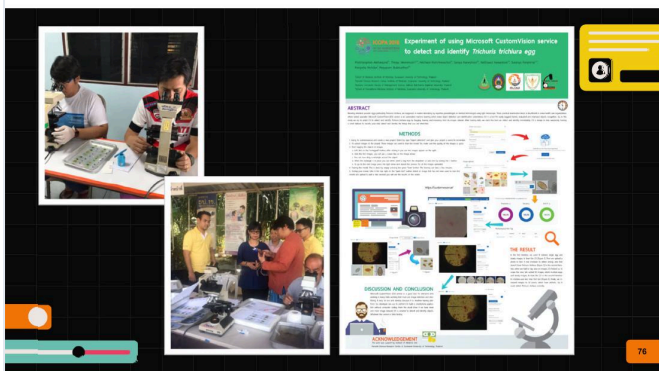
AI-Helminths: Parasite-egg detection AI (OV, MIF, Cs, Ec, AI, Tt, T spp) for DDC and NECTEC.



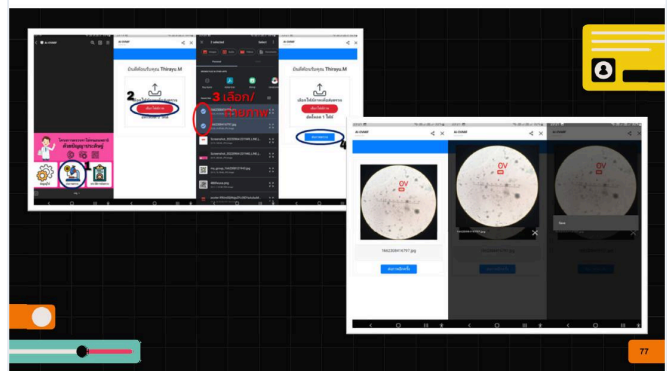
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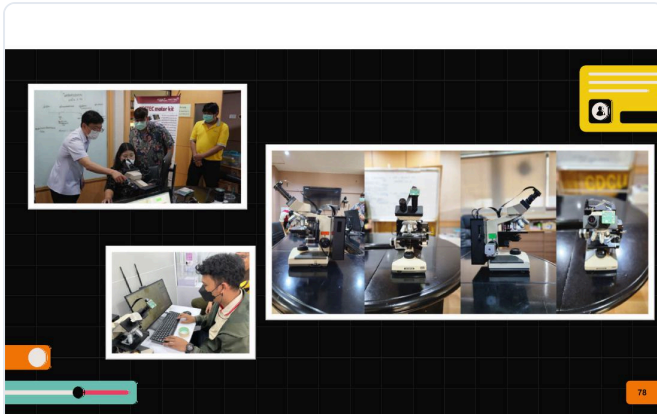
MG-OV-CCA: OV-CCA risk-screening questionnaire (LINE Liff)



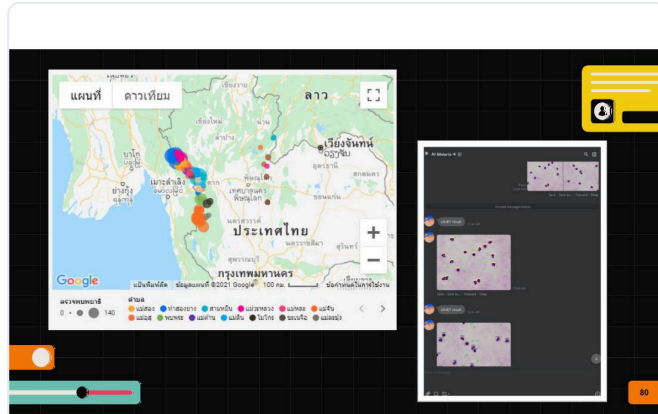
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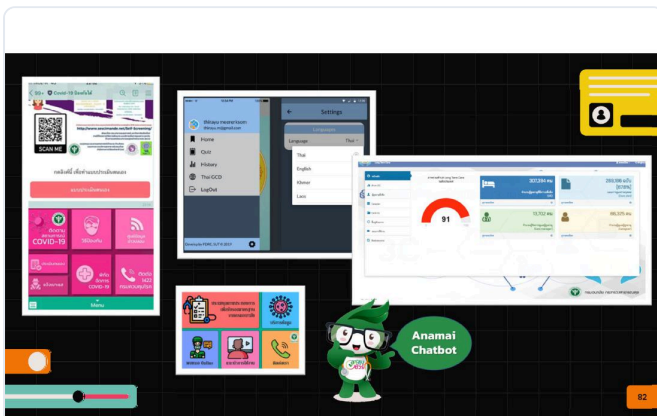
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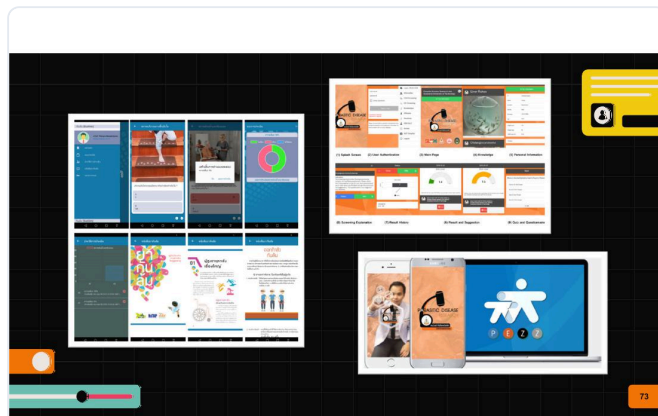
AI-Helminths: Parasite-egg detection AI (OV, MIF, Cs, Ec, AI, Tt, Tssp) for DDC and NECTEC.



AI Malaria: Computer-vision diagnosis tools for DDC. AI-Helminths: Parasite-egg detection AI (OV, MIF, Cs, Ec, AI, Tt, Tssp) for DDC and NECTEC.



3C & Long-Term Care: Web portal and GIS reporting system for the Dept. of Health. Anamai Chatbot: Rapid-response healthcare chatbot for the Dept. of Health.



PEZZ growth-analysis system & clinical apps (GAS, Bedwetting)